

Commercial Excel Capstone Project

Technical Report | 425 words

1. Problem Statement

A growing care-sector Commercial team needs reliable analysis across procurement, fleet, workforce, agency and finance data. Source files are fragmented, contain inconsistent values and require recurring reporting. The analyst must cleanse and validate data, maintain Excel financial models, track savings, investigate commercial variances and communicate concise insights while applying GDPR-first principles.

2. Solution

I built a synthetic end-to-end commercial analytics solution in Excel. Five source datasets cover 2025 procurement invoices, fleet costs, workforce costs, agency labour and finance, supported by site, supplier and savings-target lookup tables. The data deliberately contains missing values, inconsistent category casing and a duplicate invoice so data-quality controls can be demonstrated safely without using personal or confidential information.

The workbook contains Excel Tables, structured references, lookup formulas, SUMIFS/COUNTIFS, IFERROR, INDEX/MATCH, conditional formatting, data validation, charts and an executive dashboard. An Actual_Pivot sheet provides a genuine PivotTable-style analysis of spend by category and region, while Pivot_Analysis provides formula-driven summaries suitable for repeatable reporting.

3. Financial Model

The Financial_Model sheet uses a driver-based 2026 forecast rather than a static budget. A scenario selector switches between Conservative, Base and Aggressive assumptions for revenue growth, procurement savings, agency-rate inflation, fleet efficiency, payroll inflation and other operating-cost inflation. The model calculates forecast revenue, cost components, EBITDA and EBITDA margin. Savings_Tracker converts category-level savings targets into financial values and status indicators.

On the synthetic 2025 baseline, total revenue is approximately £86.5m, actual operating expenditure £60.4m and EBITDA £26.1m. Procurement spend is approximately £13.9m, with modelled category savings potential of approximately £0.78m. These figures are illustrative only and are not CareTech financial data.

4. VBA Automation

Four VBA modules are supplied: CleanseData, ConsolidateCSVs, RefreshDashboard and ExportReport. They automate data cleansing/duplicate checks, same-schema CSV

consolidation, workbook/PivotTable refresh and PDF dashboard export. The macros are designed to demonstrate intermediate VBA capability and repetitive-workflow automation.

5. Additional Technical Coverage

A Power Query/M example demonstrates text trimming, type conversion and quality-flag creation. An optional SQL script demonstrates basic aggregation and supplier-spend analysis. The project therefore also covers the desirable Power Query/M and SQL elements in the specification.

6. Outcome and Limitations

The capstone demonstrates a complete commercial workflow: ingest → cleanse → validate → analyse → model → report → automate. It directly showcases the Excel, financial-modelling, VBA, data-quality, problem-solving and stakeholder-reporting capabilities required for the Junior Data Analyst role. The dataset is synthetic and does not reproduce CareTech systems, contractual definitions or actual commercial policies; those areas would require controlled access to organisational data and agreed business definitions.